

The Effect of Epistemic Cognition Interventions on Academic Achievement: A Meta-Analysis

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Epistemic cognition involves the thinking executed as people discern what they know versus what they question, doubt, or disbelieve. Effective or adaptive epistemic cognition underlies the higher-order thinking required for life in the 21st century and has been positively correlated with academic achievement. As such, researchers have designed a number of educational interventions with the goal of developing students' epistemic cognition, but no comprehensive examination of their efficacy exists. Therefore, we conducted a meta-analysis to better understand the effect of epistemic cognition interventions upon academic achievement and what characteristics differentiated their efficacy. Twenty-six experimental and quasi-experimental studies met our inclusion criteria. Using 28 independent samples and 59 effect sizes, we found epistemic cognition interventions had a statistically significant, medium-level effect on academic achievement (Cohen's $d = 0.509$, $p < .001$). Moderator analyses revealed interventions based in guided forms of instruction and models emphasizing justification and reconciliation of objectivity and subjectivity were more successful than other interventions, suggesting several promising directions for future research and practice. Surprisingly, we found that shorter interventions were generally more successful in promoting academic performance than longer ones.

Educational Impact and Implications Statement

The proliferation of information and misinformation in the current digital age affects how people construct, justify, and use knowledge, that is, their epistemic cognition. Previous research has shown that students' epistemic cognition predicts their academic achievement. Our analyses reveal that epistemic cognition interventions, on average, have a positive influence on academic achievement. Further, epistemic cognition interventions based on source evaluation and guided inquiry instruction are particularly promising.

Keywords: epistemic cognition, academic achievement, interventions, meta-analysis, implementation

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Societies today are both technologically advanced and information-rich. These conditions have increased and continue to increase the complexity of people's daily lives (OECD, 2016). A large proportion of the global population has access to a myriad of online sources of information, but this access has brought concerns about the quality and veracity of that information. Some educators, scholars, and governments have started referring to the current situation as the post-truth era (Keyes, 2004; Strom, Haas, Danzig, Martinez, & McConnell, 2018) because untrustworthy ideas are just as easily created and propagated as accurate ones (Bromme & Beelmann, 2018; Vosoughi, Roy, &

Aral, 2018). People need to have the skills, discipline, and dispositions to carefully analyze and evaluate information (OECD, 2016; Rapp, 2016) as well as efficiently and effectively reason about knowledge and the processes involved in knowing (Sinatra, Kienhues, & Hofer, 2014) in order to contribute productively to society. This means that people need to be able to distinguish between what they know versus what they think, believe, doubt, or decide is not truth or knowledge. Such decisions in this post-truth, misinformation-rich era require adaptive epistemic cognition, which is the effective execution of thinking as "people acquire, understand, justify, change, and use knowledge" (Greene, Sandoval, & Bråten, 2016b, p. 1).

Epistemic cognition research has been ongoing for the last 50 years, but the advent of the Internet and social media has increased interest in the field (Greene et al., 2016b). Researchers have investigated the importance of epistemic cognition in academic subjects such as psychology (Hofer, 2000), mathematics (Muis, 2004), history (Maggioni, VanSledright, & Alexander, 2009), and science (Barzilai & Weinstock, 2015), as well as in distinct learning contexts such as reading for comprehension (Bråten, Britt,

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Strømsø, & Rouet, 2011), Internet searching (Kammerer, Amann, & Gerjets, 2015), and multiple source use (Bråten, Ferguson, Strømsø, & Anmarkrud, 2014). There has been a great deal of controversy regarding the best way to phrase epistemic cognition that leads to positive outcomes, including terms such as sophisticated, adaptive, availing, or effective epistemic cognition. Others have referred to terms such as epistemic competence (e.g., Murphy & Alexander, 2016) or apt epistemic performance (Barzilai & Chinn, 2018). Adjudicating the best phrasing is not the purpose of this paper, therefore we will use the term “adaptive epistemic cognition” as an omnibus, general term that can include the ones listed previously and can be taken to generally indicate when people exhibit the flexibility to adjust their epistemic cognition “to match the norms of the context in which they enact their thinking” (Greene, Sandoval, & Bråten, 2016c, p. 498) and perform successfully.

Scholars have argued that more adaptive epistemic cognition should be associated with better learning outcomes and higher academic achievement (e.g., Muis, 2004), and a recent meta-analysis (Greene, Cartiff, & Duke, 2018) provided empirical support for such a relationship, albeit with a relatively small correlational effect size. However, establishing a correlational relationship between epistemic cognition and academic achievement begs the question of whether interventions intended to improve epistemic cognition will cause subsequent increases in academic achievement. Greene and colleagues' (2018) meta-analysis did not address this point.

There is a clear need for a systematic, quantitative synthesis of research investigating the effect of epistemic cognition interventions on academic achievement. Such an analysis would extend Greene and colleagues' (2018) findings by identifying whether interventions targeting epistemic cognition can improve academic achievement, and if so what kinds of interventions seem most promising. Consequently, in this meta-analysis we addressed three research questions:

- (1) What is the overall average effect of epistemic cognition interventions upon academic achievement?
- (2) How do conceptual aspects of the interventions moderate the resultant effect size?
- (3) How do methodological factors moderate the resultant effect size?

We restricted our sample to experimental and quasi-experimental studies of epistemic cognition interventions to estimate a causal relationship. One of our goals was to identify what conceptual aspects of these interventions were associated with strongest effects, so that we could make recommendations regarding the design and development of future interventions. Further, we accounted for methodological factors (e.g., publication in peer-reviewed venues) to capture how they might affect the validity of inferences from our findings.

Epistemic Cognition Interventions

Interventions aimed at promoting epistemic cognition development were initiated soon after Perry (1970) published the seminal model in the field. Knefelkamp (1974) and Widick (1975) conducted short nonexperimental instructional interventions with college students, based on Perry's conceptualizations of relativistic

and dualistic thinking, to determine whether students' cognitive schemes could be changed. Both interventions involved challenging students through the use of alternative perspectives in class activities and resulted in most students advancing along Perry's developmental scale. Widick (1975) further investigated possible effects on academic achievement, finding some small positive impact on one of the short-essay performance outcomes.

Knefelkamp (1974) and Widick's (1975) work foreshadowed many of the aspects of the interventions that followed them. For example, subsequent research has been largely nonexperimental in nature (e.g., Carter & Yackel, 1989; Chen, 2017). Therefore, in many studies it was not clear whether changes in participants' self-reported epistemological beliefs were due to the intervention itself or aspects of the context in which the intervention was delivered (e.g., the nature of instruction, the use of diaries; Güven, Sülün, & Çam, 2014). Further, even among the various interventions that were successful in promoting more adaptive epistemic cognition or beliefs among the participants, many of the researchers implementing them (e.g., Gill, Ashton, & Algina, 2004; Kienhues, Stadler, & Bromme, 2011; Valanides & Angeli, 2005) emulated Knefelkamp's (1974) work by concentrating on epistemic cognition development resulting from the intervention, rather than measuring effects on academic achievement (Bråten, 2016).

The vast majority of experimental interventions into epistemic cognition and its effects on academic achievement have been conducted fairly recently, after the formalization of specific theories of epistemic change (e.g., Bendixen & Rule, 2004). Some scholars view epistemic change as a relatively stable process of development that takes place over longer periods of time (e.g., Kuhn, 1999), whereas other researchers have adopted a more situated approach (e.g., Elby & Hammer, 2010) in which the resources available in situ dictate the epistemic change. Developmental viewpoints have been translated into interventions, whereas situated ones have not. Nonetheless, most scholars have taken a more social constructivist approach toward conceptualizing and affecting epistemic cognition (Hofer, 2004). In addition to differences in theories framing these studies, our review of the literature also revealed that there were a number of other ways that the interventions varied.

Characteristics of Interventions

We reviewed the epistemic cognition intervention literature to identify important characteristics of the interventions. These characteristics included the instructional framework for the intervention, the theoretical model underlying the work, the level of specificity of the intervention and the achievement measure, the type of academic achievement measured, the source of intervention instruction, the educational level of the sample, the length of the intervention, and the timing of the posttest measure (see Table S1 in online supplementary material for all initial characteristics coded). We based our coding and subsequent meta-analysis on these characteristics, as well as two methodological moderators.

Instructional Framework Guiding Intervention

There has not been much research into which types of instructional models best promote epistemic cognition (Bråten, 2016).

The primary instructional frameworks used in such interventions have included constructivist instruction (e.g., focusing on the idea that learning is a personal process of building knowledge; Muis & Duffy, 2013), problem-based learning (e.g., Elen & Clarebout, 2001; Sahin, 2010a), metacognitive or cognitive scaffolding (e.g., Barzilai & Ka'adan, 2017; Heffer et al., 2015), guided inquiry (e.g., Lam & Chan, 2008; Schiefer, Golle, Tibus, Trautwein, & Oschatz, 2017), argumentation (e.g., Osborne, Simon, Christodoulou, Howell-Richardson, & Richardson, 2013; Ryu & Sandoval, 2012), and source evaluation training (e.g., Mason, Junyent, & Tornatora, 2014). For the studies included in this meta-analysis, there was great deal of variance in terms of how explicit the researchers were about the instructional framework underlying their intervention (see Table S2 in online supplementary material). Sahin (2009, 2010a, 2010b), for example, clearly framed all of his interventions as problem-based, and similarly Nussbaum, Sinatra, and Poliquin (2008) and Osborne et al. (2013) were overt about their interventions being based in argumentation. In Nussbaum et al.'s (2008) study, the treatment group received information about criteria for assessing scientific arguments, whereas the treatment groups in Osborne et al. (2013) learned more about how to engage in scientific arguments. Muis and Duffy (2013) were somewhat less explicit, but they did identify a "constructivist approach to teaching" (p. 214) as the key facet of their treatment instruction, as compared to the control condition of traditional, teacher-centered instruction. In cases such as these, coding of the study's instructional framework was relatively straightforward.

In cases in which the authors of the study did not clearly identify their instructional framework, we classified the intervention based upon its most important or salient features. For example, Mason and Scrivani (2004) described their treatment as participation in an "innovative learning environment" (p. 157), as compared to the control condition of traditional instruction. The researchers described several characteristics of the intervention, including "establishing a different classroom culture through the negotiation of new socio-mathematical norms" (Mason & Scrivani, 2004, p. 159) and "promoting and encouraging socio-cognitive interaction between students through small-group assignments and discussions" (p. 160). The driving force behind the instruction, though, seemed to be the use and discussion of "non-routine problem situations" (Mason & Scrivani, 2004, p. 161) that were designed to "stimulate learners to think about the nature of the problem and pay attention to the structure and not only surface details . . . [which] could involve students on the epistemic level of cognition" (p. 161). We decided that this description indicated that this study fit best in the problem-based learning instructional framework category. Kuhn, Arvidsson, Lesperance, and Corprew (2017), as another example, mentioned that their treatment group was involved in an extended problem-based activity, which seems to indicate that the study could have been categorized as a problem-based learning instructional framework. However, they also described their "pedagogical method . . . as one of guided inquiry" (p. 234), and it appeared to us that this aspect was more clearly at the core of their instruction, leading us to classify the instructional framework in their study as guided inquiry.

Ferguson, Bråten, Strømsø, and Anmarkrud (2013) was problematic in a different way, as we initially coded it as having a conflicting information instructional framework (i.e. treatment students read documents that presented different perspectives on

health issues). Thus, the Ferguson et al. (2013) study would have been combined with another study with a similar focus, but that second study had to be dropped from the meta-analysis after we were unable to obtain achievement data from the authors. Consequently, we had to reclassify Ferguson et al. (2013) and chose to include it with argumentation studies because the treatment students were exposed to conflicting arguments, which makes analyzing the underlying claims and reasons in these conflicting arguments more salient than reading documents with consistent arguments. Additionally, Ferguson et al. (2013) assessed participants' argumentation through their multiple source use comprehension measure and found that the treatment group performed better. In a similar manner, Olive-Taylor (2007) seemed to emphasize the main characteristic of her instruction as its interdisciplinary nature, but we did not have another study that could be included in such a category. Given that Olive-Taylor (2007) also emphasized the importance of students "constructing meaning in the classroom" (p. 8) and claimed that interdisciplinary courses allowed for "more engagement of student's prior knowledge and experience" (p. 36), we chose to classify her study's instructional framework as constructivist teaching practices. Similar investigations of the article text were conducted to classify other studies that were not explicit in their instructional framework.

Theoretical Model

There are many models of epistemic cognition, and this number continues to grow, particularly as researchers in different domains (e.g., psychology, medicine) refine models based on norms and practices in their respective fields (e.g., Weber, Inglis, & Mejia-Ramos, 2014). All of these existent models share some common origins and facets but may be categorized as either developmental, dimensional, philosophically based, or based in a specific academic discipline (Hofer, 2016). A full review of these models is beyond the scope of this study, but a brief review is warranted given the model informing an intervention may be an important predictor of its success.

Developmental models of epistemic cognition generally feature sequential stages of increasingly adaptive epistemic cognition. For example, Kuhn and colleagues' (2000) model, which remains popular and was employed by several studies included in this meta-analysis, posits that people progressively adopt increasingly sophisticated views regarding both the objective and subjective facets of knowledge. Multidimensional models, conversely, hold that a person's epistemic cognition is made up of separate beliefs, with some proponents holding that these beliefs are relatively independent (Schommer, 1990) and others proposing that they are not (Hofer & Pintrich, 1997). In either case, dimensional models share a focus on beliefs with developmental models, with the assumption that more adaptive beliefs predict better academic performance (see Sandoval, Greene, & Bråten, 2016).

Other scholars (e.g., Muis, 2008; Greene, Azevedo, & Torney-Purta, 2008; Chinn, Buckland, & Samarapungavan, 2011) have drawn from the philosophical study of epistemology to inform their models. Some of these models have focused on the justifications by which people judge to know something. Greene et al. (2008) distinguished between justification by authority and personal justification dimensions, and Bråten and colleagues added a dimension of justification by multiple sources in several empirical

and theoretical papers (e.g., Bråten et al., 2011, 2014; Ferguson, Bråten, & Strømsø, 2012). In their AIR model, Chinn and colleagues (2011, 2014) infused their model of epistemic cognition with ideas from philosophy to a greater extent than other researchers, focusing on people's engagement with epistemic aims, epistemic ideals, and reliable processes for achieving epistemic ends. To date there have been no interventions derived from the AIR model.

Researchers have also developed domain-specific epistemic cognition models (e.g., Elby & Hammer, 2010; Lee, Goldman, Levine, & Magliano, 2016). These models are based on research on learning and teaching in specific domains such as physics and history. Researchers using these models emphasize that students justify and construct knowledge in different ways depending on how they perceive the specific practices of the particular disciplines (Sandoval, 2016), rather than general beliefs about knowledge.

During coding, we combined models that had similar features to avoid small sample sizes and to add to the coherence of the analysis. Thus, the Bråten et al. model, which provided the basis for Mason et al.'s (2014) study and Ferguson et al.'s (2013) study, was combined with the Greene et al. model (e.g., Greene et al., 2008) because it was proposed by Bråten et al. (2011) using explicit links to Greene et al.'s (2008) work on multiple justifications for knowing. Ferguson et al. (2013) even overtly acknowledged this relationship stating, "we built on the philosophically grounding proposal of Greene et al. (2008), assessing a system of beliefs including different means of justifying knowledge claims" (p. 100). Muis, Bendixen, and Haerle's (2006) TIDE framework, which was used in Muis and Duffy's (2013) study, aligns with the four epistemic belief dimensions of certainty/simplicity of knowledge, justification of beliefs, source of knowledge, and attainability of truth proposed in Hofer and Pintrich's (1997) model. Additionally, Muis and Duffy (2013) explicitly mapped the TIDE framework to Hofer and Pintrich's (1997) multidimensional framework, increasing our confidence in combining these models for analysis. Barzilai and Ka'adan (2017) used the Barzilai and Zohar (2014) model to frame their study but were also explicit about this model's overlap with Kuhn's model, in that both employ the same epistemic perspectives (i.e. absolutist view, multiplist view, and evaluativist view). This relationship led us to combine these models into a joint "Kuhn/Barzilai & Zohar" code.

Other studies posed greater challenges in deciding how to combine the employed theoretical models. The model from De Corte, Eynde, and Verschaffel (2002), which helped frame Mason and Scrivani's (2004) study, shared some alignment with the same four epistemic beliefs in the Hofer and Pintrich (1997) model and therefore was coded as a study based in Hofer and Pintrich's theoretical model. Stoel, van Drie, and van Boxtel's (2017) use of the Maggioni et al. (2009) model posed a similar challenge, as Maggioni et al. (2009) made it clear that their model was based on the three distinct frameworks of Kitchener, Kuhn, and Schommer. We felt that the epistemic stances (i.e. copier, subjectivist, and criterialist), employed in Stoel et al. (2017) most clearly aligned with Kuhn's model, so we coded it as the Kuhn/Barzilai & Zohar model in the moderator analysis. In each case, particularly those in which the authors were not explicit about the theoretical model underlying their work, we endeavored to make justifiable decisions

regarding the coding of the theoretical model (see online supplementary material Table S2).

Specificity Level of the Intervention

Specificity level of the intervention. Early on, many researchers envisioned epistemic cognition as a domain-general construct with people applying their epistemic beliefs in a consistent way across diverse topics and domains (King & Kitchener, 1994; Perry, 1970). However, after Hofer and Pintrich (1997) proposed instead that these beliefs might have both domain-general and domain-specific aspects, most researchers in the field moved toward this view (e.g., Buehl & Alexander, 2001; Kuhn, Cheney, & Weinstock, 2000). Muis et al. (2006) proposed in their theory of integrated domains in epistemology (TIDE) model that the context of learning may dictate when people's domain-general or domain-specific epistemic cognition exerts greater effect. For example, different epistemic practices and types of evidence may be used across various subdisciplines of science: people may believe that chemistry is simple, whereas they believe ecology is complex. There is substantial empirical research supporting both domain-general consistencies and domain-specific differences in people's responses to a variety of epistemic cognition instruments and measures (Buehl & Alexander, 2001; Hofer, 2000).

In the last decade or so, scholars in the field have started to investigate epistemic cognition at even more specific levels, such as by topic (Barzilai & Eshet-Alkalai, 2015; Ferguson et al., 2012; Stahl & Bromme, 2007). Even within a single academic domain such as environmental studies, people may believe the study of air pollution is certain but the study of climate change is not. Some researchers have taken an even more situated view, conceptualizing epistemic cognition as fundamentally dependent upon active, in-the-moment interactions between the people, content, tasks, and resources of the specific context (Elby, Macrandar, & Hammer, 2016; Sandoval, 2012). However, we could find no experimental or quasi-experimental interventions based in this situated perspective. Nonetheless, the level of specificity of an intervention may be an important predictor of its success.

Specificity Level of the Academic Achievement Measure

Specificity level of the academic achievement measure. The specificity level of the achievement outcome may moderate the effect of an epistemic cognition intervention as well. Domain-general measures, like GPA, consist of many forms of knowledge across various subjects and therefore may not be sensitive to domain- or topic-specific epistemic cognition interventions. Domain-specific measures, like a final exam, are subject-specific but still cover many different topics. Topic-specific measures might be most sensitive to a very focused epistemic cognition intervention, but less so with domain-general interventions aimed at fostering adaptive beliefs or thinking across topics or domains. Finally, the alignment of the level of specificity between the intervention and the academic achievement measure may be an important predictor of efficacy; therefore we included this factor in our analyses as well.

Type of Academic Achievement

One reason that epistemic cognition has been the focus of much study in recent years is because of its importance in higher-order cognitive processing (Greene & Yu, 2016; Sinatra et al., 2014). It seems less likely that epistemic cognition has a large role in the acquisition of declarative knowledge (i.e. memorized knowledge of facts; Phye, 1997) or procedural knowledge (i.e. knowledge of how to do or accomplish something; Krathwohl, 2002). However, understanding the nature of knowledge and reliable processes for obtaining and evaluating it seems critical for conceptual knowledge (i.e. “understanding of the principles that govern a domain and of the interrelations between pieces of knowledge,” Rittle-Johnson & Alibali, 1999, p. 175), and epistemic cognition and conceptual change appear to be strongly interdependent (Lombardi, Nussbaum, & Sinatra, 2016; Sinatra et al., 2014). Adaptive epistemic cognition also appears to be related to argumentative knowledge or argumentative reasoning (Duncan & Chinn, 2016; Herrenkohl & Cornelius, 2013; Iordanou, Kendeou, & Beker, 2016). In fact, in some of the literature, argumentation and epistemic cognition are so intertwined as to be almost indistinguishable from one another (Bråten, 2016). Therefore, a meta-analysis of the efficacy of epistemic cognition interventions should account for the type of academic achievement outcome measured.

Achievement Measure Developer

Muis, Trevors, and Chevrier (2016) pointed out the importance of academic assessments aligning with the “pedagogical approaches to foster epistemic change” (p. 353). Whereas Muis et al. (2016) stressed the importance of teachers embracing that need for alignment, meta-analyses must also account for potential variance in effect sizes by the developer of the achievement measure. Researcher-developed measures may be more sensitive to changes due to the intervention than standardized or general GPA measures, but issues of external validity must also be considered (Ahn, Ames, & Myers, 2012).

Source of Intervention Instruction

As with the developer of the achievement measure, the source of the intervention instruction may also be expected to moderate the results. Teachers with established relationships with their students are likely to demand greater attention from them. Researchers could be expected to implement their interventions with greater fidelity. Therefore, teams of researchers and teachers might be expected to have the greatest impact when working together to implement the intervention. There are also self-administered interventions in which participants complete a learning module on a computer or read through an instructional handout. Therefore variance in the source of the intervention may relate to its efficacy.

Educational Level

Perry (1970) expected incoming college students to be fairly undeveloped with regard to their epistemic cognition. Many scholars have proposed that both biological maturation and educational experience are necessary for the development of epistemic cognition, with little expectation of such growth prior to the college years (e.g., King & Kitchener, 1994). Chandler, Hallett, and Sokol

(2002) argued against such assumptions, and empirical studies, both observational and experimental, have indeed shown earlier development, as well as significant interindividual variability. For example, elementary school students, assumed to have “naïve” beliefs within the early developmental models, have been shown to be capable of sophisticated or adaptive epistemic cognition (e.g., Conley, Pintrich, Vekiri, & Harrison, 2004; Metz, 2011). Researchers have found middle-school students can have a diverse and complex epistemic cognition (e.g., Mason, Boldrin, & Ariasi, 2010). Greene et al. (2018) showed that the two highest correlations between adaptive epistemic cognition and academic achievement were for elementary and middle-school students, further suggesting significant variability among children at those ages.

Length of Intervention

Many proponents of the developmental models postulate that epistemic cognition develops gradually over time and is generally fostered by social and academic experiences (Weinstock & Zviling-Beiser, 2009). However, epistemic cognition interventions are designed to advance this process (e.g., Ferguson et al., 2012). What is not clear is how long interventions have to be to have lasting effects on academic achievement. In our sample, some were as short as a single session, whereas others lasted a semester or longer.

There were complications to analyzing the length of intervention as a moderator. Some interventions were given in multiple sessions, and others were not. Additionally, the articles were generally clear in how long studies lasted, but it was not always evident what the exact length or dosage of the enacted intervention was. These uncertainties and complications led us to create fairly broad, open categories for the moderator analysis.

Timing of the Posttest Achievement Measure

One of the goals of an epistemic cognition intervention is to affect academic achievement both in the short- and long-term (Bråten, 2016). Such effects are often captured via immediate and delayed posttest measures. It may also be the case that epistemic cognition interventions only affect thinking after some period of time, as students internalize adaptive beliefs and practices.

Methodological Moderators

We identified two potentially relevant aspects of the study methodology that might affect efficacy. First, we wished to compare whether the average effect varied across true- and quasi-experimental research designs. We also wanted to examine any differences between studies published in peer-reviewed journals as compared to those that appeared in the “gray” literature such as conference presentations. These two factors might moderate the size of the effect size estimates derived in the analysis.

Method

Inclusion Criteria

We had several primary criteria for studies to be included in the meta-analysis: (a) the article clearly indicated the intervention was

targeting epistemic cognition or epistemic beliefs, (b) the research design was experimental or quasi-experimental, (c) a measure of academic achievement was collected after the intervention, (d) the effect size (Cohen's d) for each treatment and control group was reported or enough information was provided to compute the effect size, and (e) the study was published in English.

Literature Search Strategy

Figure 1 displays the search and inclusion process used to identify relevant literature. We started with 24 papers, collected in a previous study, that we believed were likely to be informative for this study. Then, we systematically searched 13 electronic databases: Google Scholar, PsycINFO, ERIC, Education Full Text, Articles+, Academic Premiere, and ProQuest Dissertations &

Theses Global, as well as the Social Sciences Citation Index, Arts & Humanities Citation Index, Conference Proceedings Citation Index—Social Science & Humanities, Book Citation Index—Social Sciences & Humanities, and the Emerging Sources Citation Index, which are all parts of the Web of Science Core Collection. We used a wide variety of terms to locate all relevant studies (e.g., epistemic cognition, personal epistemology, epistemological beliefs), as the terminology in the field has changed and expanded over time, and we wanted to include all pertinent studies regardless of the employed terminology. Table 1 provides an overview of the electronic database search procedure, including search terms.

Several other methods were also used to identify related literature. We reviewed the content and references of seven prominent books on epistemic cognition (Bendixen & Feucht, 2010; Brown-

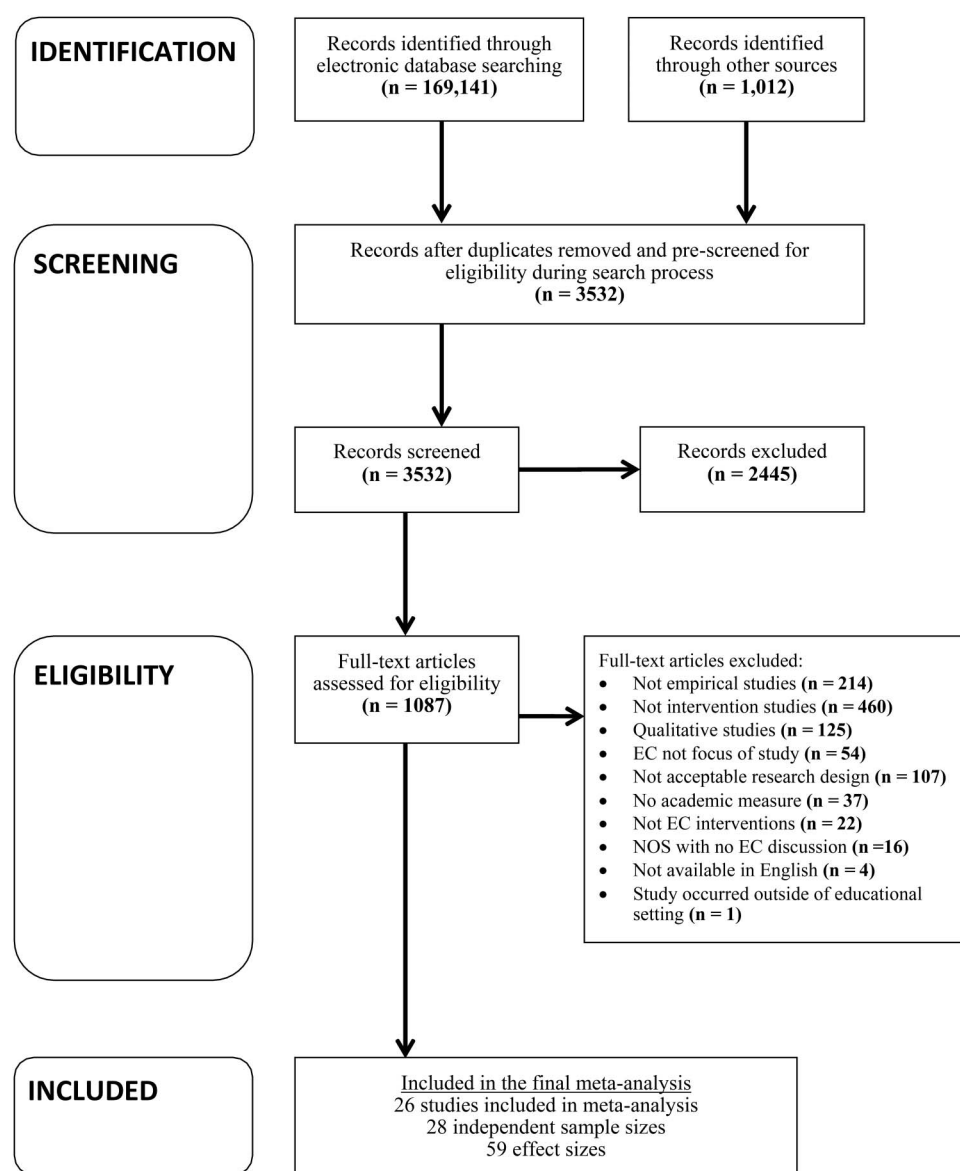


Figure 1. Flow chart of literature search and exclusion process.

Table 1
Overview of Literature Search Procedure for Electronic Databases

Literature search	Search terms	Electronic databases
November and December 2017	"epistemic," "cognition," and "intervention"; "epistem*," "belie*," and "intervention"	ERIC, PsycINFO, Education Full Text, Web of Science, Citation Index
December 2017	"epistemic," "beliefs," and "intervention"	Articles +
December 2017	"epistem" and "belie*" and "change,"; "personal epistemology" and "intervention"; "epistemic," "beliefs," and "intervention"; "epistemic," "cognition," and "intervention"	Academic Search Premier, ERIC, Education Full Text, PsycINFO
December 2017	"epistemic," "beliefs," and "change"	Google Scholar
January 2018	"epistem*" and "intervention"	Academic Search Premier, ERIC, Education Full Text, PsycINFO
February 2018	"epistem*," "belie*," and "intervention"	ERIC, Education Full Text, PsycINFO
March 2018	"epistem*," and "control"; "epistem*," and "experimental"	Academic Search Premier, ERIC, Education Full Text, PsycINFO
March 2018	"epistemic," "cognition," and "intervention"; "epistemic," "beliefs," and "intervention"; "personal," epistemology," and "intervention"	ProQuest Dissertation & Thesis Abstracts, Web of Science, Citation Index
June 2018	"epistemic," "beliefs," and "intervention" and "epistemic," "beliefs," and "intervention" (restricted to 2017–2018 publications)	Academic Search Premier, ERIC, Education Full Text, PsycINFO

Note. The asterisk was used as a wildcard symbol to capture variations of a term.

lee, Schraw, & Berthelsen, 2011; Fives & Gill, 2015; Greene, Sandoval, & Bråten, 2016a; Hofer & Pintrich, 2002; Khine, 2008; Schraw, Brownlee, Olafson, & Vanderveldt, 2017). We also examined the references of an additional 75 sources including seminal epistemic cognition intervention papers (e.g., Gill et al., 2004; Valanides & Angeli, 2005) and relevant epistemic cognition reviews (e.g., Muis, 2004; Sandoval et al., 2016). Additionally, the table of contents of every publication of *Journal of Educational Psychology* and *Contemporary Educational Psychology* from 1996 through January of 2018 were scanned to ensure no relevant studies were missed.

In order to reduce publication bias and include all applicable studies, we searched the "gray" literature for unpublished studies. We first searched online conference repositories for the stem "epistem*." Studies listed in the American Educational Research Association online paper repositories from 2010 to 2017 were reviewed. All available electronic conference programs for the American Psychological Association Division 15 (2014–2017), the National Association for Research in Science Teaching (2010–2017), and the conferences on Personal Epistemology and Learning (PEL, 2009, 2016) were examined as well. We also searched available Proceedings of the International Conference of the Learning Sciences (2010, 2012, 2014, 2016), and programs from the International Conference on Conceptual Change (2012, 2014, 2016), the World Conference on Educational Services (2009–2013), the Association for Psychological Science (APS, 2013–2017), and others. If full articles were not readily available but seemed to be aligned with the purpose of the meta-analysis, we contacted the first authors and requested articles. Finally, we requested unpublished articles by emailing 42 researchers who have published multiple studies on epistemic cognition or epistemic beliefs in the last 20 years and sent a general e-mail request through a listserv (epistemiccognition@

listserv.unc.edu). Twenty-two of these researchers responded, but none of them had any pertinent data to share.

The initial search identification process resulted in over 170,000 documents. Prescreening (i.e. scanning for relevance to the topic of epistemic cognition) and removing duplicates narrowed this down to 3,532 documents to screen for inclusion in the meta-analysis. The first author examined these documents for eligibility by reading the titles, abstracts, and keywords and eliminated 2,445 articles. We obtained full-text copies of the remaining 1,087 papers, which the first author then examined more fully by reading the studies' methods, procedures, and data collection sections. Borderline cases were shared with the third author for further discussion. Ultimately, these steps led to the inclusion of 26 studies, comprised of 20 peer-reviewed journal articles, four dissertations, and two conference proceedings.

Coding Procedure

Three studies were coded for a priori categories by all three authors independently. Then the authors met to discuss possible modifications to the coding scheme and confer on any difficulties in coding. The first two authors then created the coding manual and subsequently coded five more studies independently to measure their consistency. After the rest of the studies were coded by the first two authors, the overall interrater agreement was .94. All discrepancies were discussed and resolved with the third author acting as an arbiter when necessary. All of the studies and the coded information can be found in the Table S2 in online supplementary material, which has several codes not discussed in the paper because there was little variance across studies on these codes.

Statistical Procedures

Effect size estimation. We used Cohen's d as a measure of effect size. For studies that did not directly report Cohen's d values, either (a) reported means and standard deviations for the control and treatment groups or (b) chi-square values with one degree of freedom were used to calculate Cohen's d values using a converter (Stangroom, 2019). Additionally, one author was contacted to retrieve relevant data that was not reported in the published article (Olive-Taylor, 2007). When the effect size was reported such that a more negative score indicated higher achievement (e.g., Nussbaum et al., 2008), we reversed the sign of the effect directions to ensure that they were all scaled to indicate higher positive scores meant greater achievement.

Identification of statistical outliers. Before conducting a full analysis, we generated histograms to assess the normality of the distribution of the effect sizes. These plots indicated a high degree of kurtosis, which called assumptions of normality into doubt. We ran a Grubbs' (1950) outlier test, which revealed one outlier ($d = 3.2$ from Kuhn et al., 2017) at the $p < .05$ significance level (two-tailed). There are three major options for dealing with outliers: (a) including them intact, (b) Winsorizing them (i.e. using an attenuated value instead of the original value; Lipsey & Wilson, 2001), or (c) trimming the value from the analysis. Including the outlier did not seem reasonable given that it would affect assumptions about normality that are critical to a meta-analysis. Eliminating the value completely also seemed unreasonable given that

there was no evidence that the effect size was an error (Ghosh & Vogt, 2012). We therefore decided to Winsorize the datum by adjusting it to the next closest effect size ($d = 2.6$). New histograms and analyses were conducted with the adjusted value, and the adjusted data were deemed to exhibit acceptable assumptions of normality and were consequently used throughout the rest of the analysis.

Publication bias. Despite our best attempts to be as thorough in our search as possible, we recognized the potential for publication bias, including the possibility of unpublished results due to nonstatistical findings (i.e. the "file-drawer" problem; Carson, Schriesheim, & Kinicki, 1990). To explore this possibility, first we created a funnel plot (see Figure 2). The funnel plot was asymmetrical, indicating the possible presence of publication bias (Egger, Smith, Schneider, & Minder, 1997). Imputing missing studies seemed to significantly change the appearance of the funnel plot, further supporting the possibility of publication bias. Additionally, the Duval and Tweedie trim-and-fill and Egger's regression test (Egger et al., 1997) were run, and each further indicated that publication bias is likely affecting the results of this analysis.

Overall effect analyses. Comprehensive Meta-Analysis software (CMA, Version 3.3.070; Borenstein, Hedges, Higgins, & Rothstein, 2014) was used to conduct all meta-analyses, including moderator analyses. One issue that we had to handle carefully was the weighting based on the sample sizes of the included studies. Several of the studies (e.g., Angeli, 1999; Kim & Keller, 2010)

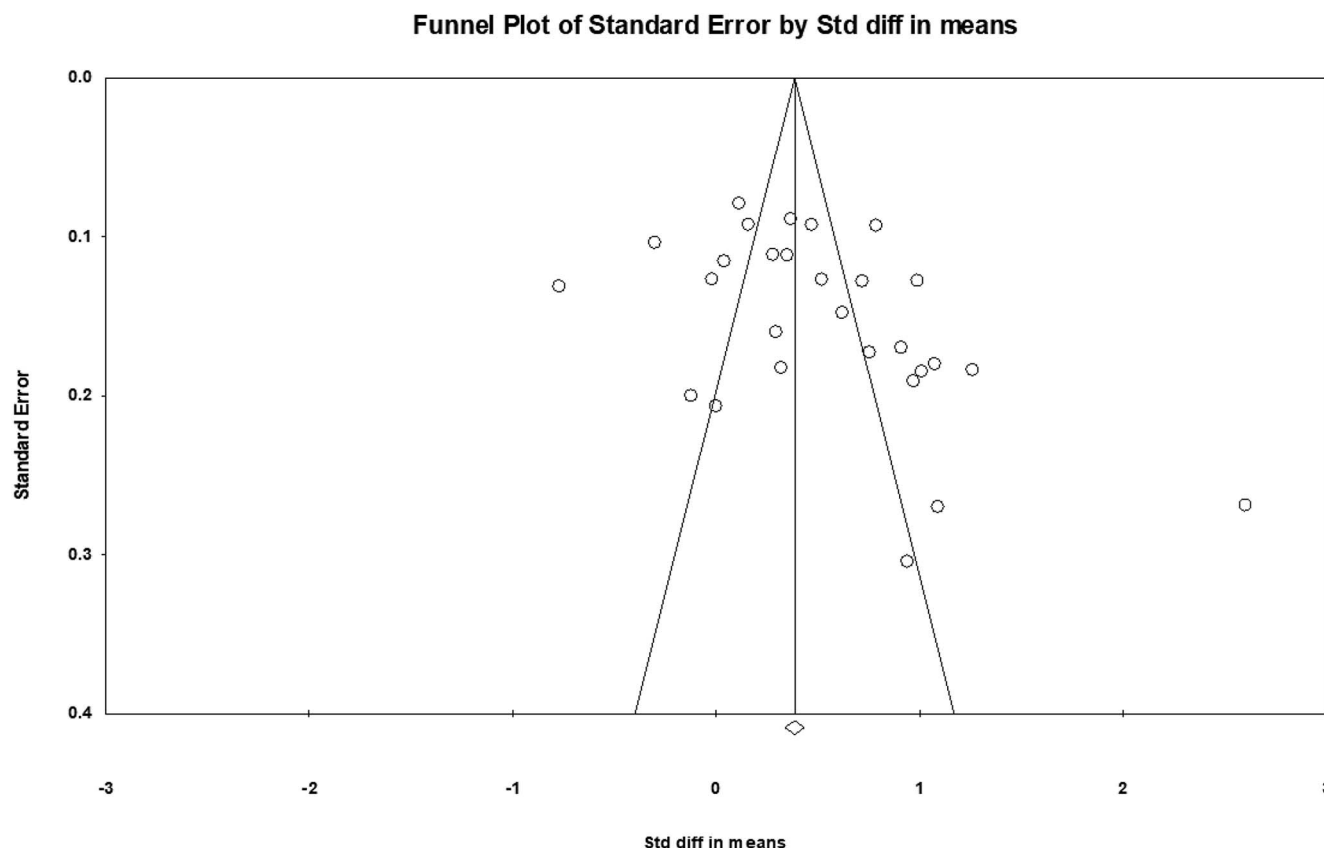


Figure 2. Funnel plot of standard error by fisher's Z .

involved additional experimental groups that were deemed to be outside the purview of epistemic cognition interventions and were not used in the meta-analysis. Therefore, we had to adjust the sample sizes from the overall sizes reported in the studies accordingly. Additionally, several of the studies had different numbers of participants related to different academic outcomes (e.g., Schiefer et al., 2017) or disparate numbers in cases where they gave both immediate and delayed assessments of academic achievement (Hefter et al., 2015). For each individual moderator analysis, we used the most appropriate sample size so as not skew the results (e.g., in a study with two posttests, if 60 students participated in the immediate testing, but only 58 students took the delayed test, then when conducting a moderator analysis with immediate testing we treated the sample as having 60 students, whereas for an analysis of effects on delayed tests we used the sample size of 58).

Additionally, in the full analysis of all effect sizes, there were several instances where a study had multiple relevant dependent values, which can be handled in several ways (Tanner-Smith, Tipton, & Polanin, 2016). In this study, we ran two overall effect size estimate models, one in which we assumed that the study as whole could be used as the unit of study (i.e. that the dependent values were perfectly correlated with each other and could therefore be averaged) and one in which we treated the values as if they were uncorrelated and independent (i.e. each effect size treated as a separate unit of study, leading to 59 effect sizes). The two overall effect size estimates, using the random effects model, differed by only .014, and the estimates in both cases were statistically significant at the $p < .001$ level. This small difference and the statistical significance of each model led us to conclude that no further adjustments for dependent correlations were necessary, and we conducted the overall effect size analysis averaging the within-study values. This led to a total of 28 distinct samples, as two of the 26 studies involved two groups of participants.

Moderator analyses. In the case of the moderator analyses (e.g., time of posttest), we adjusted for dependent correlations by taking the mean within levels of the moderator. This “shifting unit of analysis” approach (Cooper, 2017) was employed because we had multiple levels of moderators within some studies (e.g., Barzilai & Ka’adan, 2017; Lam & Chan, 2008). This approach did lead to some of the moderators being calculated as if there were more than 28 distinct samples and to slightly different numbers of participants, as the same groups of students had to be treated independently in these analyses. For example, for the Achievement Measure Developer moderator, Lam and Chan (2008) used two different types of measures, one of which was categorized as “researcher developed” and one which was a “standardized test.” Similarly, Shekoyan (2009) had different measures, some which

were coded as “researcher developed” and others that were “teacher developed.” Separating these groups of different measures meant that for the test of that particular moderator, we had to treat these two studies as if they each had two distinct samples, bringing the total number of samples to 30.

In addition to looking at each moderator individually, we also attempted to conduct metaregression analyses with various methods of combining levels of the moderators. However, these analyses failed, likely due to the limited number of studies. Even metaregressions with a limited number of predictors proved uninformative (i.e. only very simple models that did not explain any variance could even be successfully conducted). Therefore, we do not report any of these metaregression analyses. Nonetheless, we feel the individual moderator analyses yielded important information for the field.

Results

The dataset included 26 articles, 28 distinct samples, with 59 effect sizes, and 4,373 unique participants. The guiding research question focused on the effect of epistemic cognition interventions on academic performance. This examination was supplemented by the analysis of several conceptual and methodological moderators. I-Square values indicated the percentage of the variance that can be attributed to actual between-groups variance. For moderator analyses, determination of whether there was statistically significant variance across moderator levels was determined by examining the mixed effect Q -statistic.

Overall Effect Analysis

Our meta-analysis produced an overall average effect size of $d = .509$, which was statistically significant at the $p < .001$ level (see Table 2). Therefore, these analyses revealed a medium (Cohen, 1992) and statistically significant overall effect of epistemic cognition interventions on academic achievement. Over 90% of that variance in effect sizes was due to true heterogeneity in the sample ($I^2 = 92.738$). We conducted moderator analyses to examine the sources of heterogeneity.

What Features of Epistemic Cognition Interventions Affect Their Efficacy?

Instructional framework guiding intervention. Four types of frameworks (i.e., cognitive, epistemic cognition, or metacognitive scaffolds, constructivist teaching practices, guided inquiry, and source evaluation) had statistically significant effect sizes (see

Table 2
Overall Effect Size Obtained Using Effect Size From All Included Studies

Model	k	ES	95% CI		Z -value	Test of heterogeneity		
			Lower limit	Upper limit		Q	df	I^2
Fixed effects	28	.372***	.324	.420	15.149	370.964***	27	92.722
Random effects	28	.509***	.326	.692	5.452			

Note. k = Number of independent samples; number of effect sizes = 59; total $N = 4373$. CI = confidence interval. Outlier has been Winsorized to two standard deviations from the mean.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 3
Instructional Framework Guiding Intervention

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Argumentation	1017	5	.134	-.157	.425	.904	27.562***	4	
Cognitive, epistemic cognition, or metacognition scaffolds	341	4	.663**	.228	1.098	2.988	24.496***	3	
Constructivist teaching practices	690	3	.479*	.082	.877	2.366	17.523***	2	
Guided inquiry	893	7	.808***	.313	1.302	3.198	101.035***	6	
Problem-based learning	1250	7	.340	-.080	.761	1.586	115.279***	6	
Source evaluation	182	2	.800***	.625	.974	8.992	.232	1	
Within levels							286.125***	22	
Between levels ^a							17.491**/84.839***	5	
Overall	4373	28	.600***	.477	.723	9.563	370.964***	27	

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 3). Two of the more popular types of interventions, argumentation and problem-based learning, had much lower average effect sizes that were not statistically significant.

Theoretical model. As shown in Table 4, the estimated effect sizes by theoretical model were fairly variable. The overall effect size was statistically significant, as were the results from several of the individual (or combined/collapsed) models, including the Kuhn/Barzilai and Zohar category and the Hofer and Pintrich model with the two largest effect sizes. The results for studies using the Hammer model and the Schommer-Aikins model were not statistically significant.

Specificity level of the intervention. Studies at all three levels had statistically significant average effect sizes, though the domain-general interventions and the topic-specific interventions had much higher values than the domain-specific interventions. However, their confidence intervals all overlapped, and the Q statistic did not indicate statistically significant differences between levels, as seen in Table 5.

Specificity of academic achievement. Similar to the specificity level of the intervention, the specificity level of the achievement measure also yielded statistically significant average effect sizes at all three levels (see Table 6), and the domain-general and topic-specific results were much greater than the domain-specific results. In the case of this moderator, the Q statistic did reveal that the differences between groups were statistically significant.

Alignment of specificity level. The six studies that had unaligned levels of specificity between the intervention and the achievement measure (e.g., a domain-specific intervention with a domain-general achievement measure) had almost twice the average effect size as the studies that were aligned, with no overlap in the 95% confidence interval (see Table 7). Both average effect sizes were statistically significantly different than zero.

Type of academic achievement. The argumentation, conceptual knowledge, and declarative knowledge categories of the type of achievement moderator all had statistically significant effect sizes (see Table 8). Studies that measured general knowledge or procedural knowledge did not yield statistically significant effect sizes, though there was only one effect size (from Schiefer et al., 2017) related to procedural knowledge.

Academic achievement developer. The achievement developer moderator analysis results can be seen in Table 9. The Q test for between-groups heterogeneity was not statistically significant, but with a few more studies might have reached this level. Researcher-developed assessments were the most frequently used, had the highest average effect size, and the average effect size was statistically significant. The results on standardized tests and teacher-developed assessments were much lower and were not statistically significant. GPA was statistically significant, but only used in one study (i.e. Olive-Taylor, 2007).

Table 4
Theoretical Model

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Greene/Bråten	557	4	.482*	.013	.951	2.106	29.369***	3	89.785
Hammer	530	3	.482	-.883	1.846	.692	104.029***	2	98.077
Hofer & Pintrich	1736	10	.625***	.251	1.000	3.275	163.254***	9	94.487
Kitchener	489	3	.391**	.093	.689	2.573	9.405**	2	78.734
Kuhn/Barzilai & Zohar	396	5	.633**	.293	.973	3.653	27.605***	4	85.510
Schommer-Aikins	665	3	.185	-.124	.495	1.176	12.015**	2	83.354
Within levels							345.677***	22	
Between levels ^a							4.929/25.287***	5	
Overall	4373	28	.440***	.287	.593	5.629	370.964***	27	92.722

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 5
Specificity Level of Intervention

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Domain-general	879	5	.685***	.527	.844	8.485	7.415	4	46.056
Domain-specific	2668	16	.360**	.124	.596	2.987	199.772***	15	92.491
Topic-specific	826	7	.722**	.236	1.207	2.911	113.313***	6	94.705
Within levels							320.500***	25	
Between levels ^a							5.308/50.464***	2	
Overall	4373	28	.594***	.467	.721	9.169	370.964***	27	92.722

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Source of intervention. The source of the intervention moderator analysis (see Table 10) yielded statistically significant effect sizes for all groups. Teacher-implemented interventions had essentially the same effect size as those that were implemented by researchers. The highest effect sizes came from interventions either implemented by a team of researchers and teachers or those that were self-administered, although overall there was no statistically significant between-groups difference.

Educational level. We originally examined K-12 students separated into the categories of elementary, middle, and high school, but the three confidence intervals overlapped to a large extent, suggesting there was no need to separate them. The educational level with the highest effect size was adult students (both graduate students and those no longer in formal education), as seen in Table 11, though this was only based on two studies.

Length of intervention. Table 12 contains the analysis results for the collapsed length of intervention moderator category. If the intervention was longer than a single session but took place in one week or less, we coded it as “short term.” An “intermediate” code was applied to studies that had interventions lasting over a week but less than or equal to a month. Those studies that were greater than a month and less than or equal to three months in length were coded as “long-term” and studies beyond that length were all combined into a “semester long” category (though Osborne et al., 2013 extended greatly beyond this length to two years).

The mixed effects model estimated effect sizes that were statistically significant for the three shorter time-frame categories (i.e. single session, short-term, and intermediate) but were not for the two categories of longer-term interventions (i.e., long-term and semester). The semester category had the most studies and greatest number of participants, but overall the lowest effect size.

Time of posttest achievement. The time of the achievement posttest moderator yielded some unexpected results (see Table 13). Both immediate measures, given at the end of the intervention, and delayed measures had statistically significant effect sizes, but the average effect size of the delayed measures was more than two times larger than the average effect size of the immediate measures of achievement, a statistically significant difference.

How do Methodological Features Moderate the Effect Size?

Research design. In the moderator analysis results for research design (see Table 14), the average effect size for studies with experimental and quasi-experimental designs were similar, though both were lower than the effect size for quasi-experimental studies with equated groups. All three designs had statistically significant effect sizes, though, and their confidence intervals all overlapped, indicating no statistical differences based on the particular research design.

Table 6
Specificity Level of Academic Achievement

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Domain-general	664	3	.715***	.388	1.042	4.282	7.653*	2	73.865
Domain-specific	2633	15	.277*	.048	.506	2.368	162.994***	14	91.411
Topic-specific	1243	12	.696***	.383	1.008	4.367	144.221***	11	92.373
Within levels							314.868***	27	
Between levels ^a							6.814*/63.535***	2	
Overall	4540	30 ^b	.494***	.333	.654	6.016	378.403***	29	92.336

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

^b Lam and Chan (2008) had to be separated into two different “studies” as one achievement measure was coded as domain-specific and another was coded as topic-specific. Nussbaum, Sinatra, and Poliquin (2008) had to be separated into one study that was composed of five effect sizes coded as domain-specific, and a second study had one effect size coded with an achievement measure that was topic-specific. This led to those participants in the study being counted twice as unique.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 7
Alignment of Level of Specificity of Intervention and Achievement Measure

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Aligned	3978	23	.426***	.226	.627	4.169	303.719***	22	92.756
Not Aligned	474	6	.808***	.635	.981	9.159	8.537	5	41.431
Within levels							312.256***	27	
Between levels ^a							7.986**/58.756***	1	
Overall	4452	29 ^b	.645***	.514	.776	9.658	371.012***	28	92.453

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

^b Lam and Chan (2008) had to be separated into two different "studies" because one achievement measure was deemed to be aligned with the level of the specificity and another was coded as not aligned. This led to those participants in the study being counted twice as unique.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Peer-review status. As expected, the peer reviewed studies did have higher estimated effect size than the ones that were not peer reviewed, though both types of study had statistically significant effect sizes (see Table 15). The confidence intervals overlapped, indicating that there was no statistically significant difference between the two categories.

Discussion

Given contemporary interest in higher-order educational outcomes, such as critical thinking in the post-truth era (OECD, 2016; Rapp, 2016), researchers and practitioners have been increasingly focused upon epistemic cognition (Sandoval et al., 2016), how it relates to academic achievement (Greene et al., 2018), and whether interventions can be developed to increase effective epistemic cognition (Bråten, 2016). We used meta-analytic methods to examine the current experimental and quasi-experimental literature on epistemic cognition interventions and found across 28 samples, 59 effect sizes, and 4,373 participants that these initiatives, on average, led to statistically significant, medium-sized gains in academic achievement. This effect size would be above the average effect size of educational interventions as determined by Hattie (2015) and would be close to the top 50 of the 195 different types

of interventions studied. This finding suggests there is promise in continued efforts to bolster students' epistemic cognition and subsequent academic achievement. However, there was statistically significant variability in the average effect size estimate, indicating the importance of investigating both conceptual and methodological moderators.

We grouped each intervention into one of six types of instructional framework and found that those focused on explicit instruction or some form of guided inquiry (e.g., Kuhn et al., 2017) outperformed problem-based learning and argumentation interventions (e.g., Mason & Scrivani, 2004; Osborne et al., 2013). Researchers have suggested that engaging students with contrasting perspectives is critical for facilitating epistemic change (Kienhues, Ferguson, & Stahl, 2016). Source evaluation interventions expose students to sources with multiple, diverging perspectives on a topic and directly teach students strategies on how to evaluate these sources.

It was somewhat surprising that argumentation interventions (e.g., Nussbaum et al., 2008) were not more strongly associated with academic achievement. Argumentation interventions vary widely (Cavagnetto, 2010) both with regard to how explicit the instruction is and the aim of that instruction, as some interventions

Table 8
Type of Academic Achievement

Mixed effects analysis	N	k	ES	95% CI		Z-value	Tests of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Argumentation	477	5	1.047***	.481	1.614	3.622	69.849***	4	94.273
Conceptual knowledge	2455	19	.557***	.323	.791	4.665	215.295***	18	91.639
Declarative knowledge	261	3	.494***	.244	.743	3.874	2.807	2	28.755
General knowledge	1529	5	.127	-.383	.637	.488	90.155***	4	95.563
Procedural knowledge	100	1	.140	-.257	.537	.692	0.000	0	0.000
Within levels									
Between levels ^a							378.106	28	
							8.940/76.981***	4	
Overall	4822	33 ^b	.477***	.332	.622	6.456	455.088***	32	92.968

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

^b Several studies (Kuhn, Arvidsson, Lesperance, & Corprew, 2017; Lam & Chan, 2008; Mason, Junyent, & Tornatora, 2014; Nussbaum et al., 2008; Schiefer, Golle, Tibus, Trautwein, & Oschatz, 2017) used multiple achievement types that were coded differently. The related effect sizes had to be separated to conduct the moderator analysis. This led to those participants in the studies being counted twice as unique.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 9
Achievement Measure Developer

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
GPA (official)	490	1	.470***	.289	.651	5.083	0.000	0	0.000
Researcher developed	2429	20	.651***	.456	.846	6.539	189.986***	19	89.999
Standardized test	915	4	.304	-.208	.816	1.164	39.123***	3	92.332
Teacher developed	789	5	.012	-.449	.473	.052	47.768***	4	91.626
Within levels							276.877***	26	
Between levels ^a							7.271/96.466***	3	
Overall	4623	30 ^b	.500***	.376	.624	7.918	373.343***	29	92.232

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

^b Two studies (Lam & Chan, 2008; Shekoyan, 2009 — second group of students) used multiple achievement measures that were coded differently. The related effect sizes had to be separated to conduct the moderator analysis. This led to those participants in the studies being counted twice as unique.

* $p < .05$. ** $p < .01$. *** $p < .001$.

primarily focus on teaching students the form and structure of arguments and how to construct their own arguments whereas others focus on qualities of strong evidence and support. In the four such studies included in this meta-analysis, this variety was evident. For example, Ferguson et al. (2013) had the students in their experimental group read conflicting accounts about sun exposure and health, whereas the control group read consistent accounts, but no explicit directions on argument evaluation were provided. Stoel et al. (2017) had participating teachers model and scaffold causal connections for students in their historical discussions, but they also were not explicit in their argumentation instruction. Nussbaum et al. (2008) and Osborne et al. (2013), on the other hand, were much more explicit in teaching students the structure of arguments and how to make strong and effective arguments. The heterogeneity of argumentation intervention efficacy suggests the need for further research into how these interventions can best be designed. In addition, given the strong theoretical alignment between epistemic cognition and argumentation (Iordanou et al., 2016), research into why these argumentation interventions have yielded weaker effects than interventions built on other instructional frameworks is also warranted.

Ultimately, several of the instructional frameworks only appeared in a small number of studies, so definitive conclusions were elusive. Nonetheless, these findings do suggest that researchers and educators should consider continued development of interven-

tions that provide learners with instruction and support on how to enact effective epistemic cognition before fading that support. Such interventions would align well with Barzilai and Chinn's (2018) recent conceptualization of apt epistemic performance, wherein they argued that people must be taught to adaptively and reflectively enact epistemic cognition to achieve competence.

In terms of the models of epistemic cognition underlying the interventions, interventions based in philosophical discussions of justification (e.g., Kammerer et al., 2015; Mason et al., 2014) and the nature of knowledge and knowing (e.g., Huang, Ge, & Eseryel, 2017) were most promising. Again, the relatively small number of studies involving other models (e.g., Hammer, Schommer-Aikins) must qualify interpretations, but these results do align with criticisms of models focused on factors such as the nature of learning (cf. Hofer & Pintrich, 1997; Sandoval, 2009) and highly situated models. It is also important to note that Chinn and colleagues' (e.g., Barzilai & Chinn, 2018; Chinn et al., 2011, 2014) more situated AIR model did not figure in this analysis, therefore it seems necessary to continue to investigate interventions based in this work.

Debates about the proper level or levels of specificity of epistemic cognition (e.g., domain-general vs. domain-specific or topic-specific) persist in the field, but our findings suggest the utility of interventions targeting all three levels, supporting views such as those included in the TIDE model (Merk, Rosman, Muis, Kelava,

Table 10
Source of Epistemic Cognition Intervention

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Researcher	967	7	.422**	.133	.710	2.863	41.106***	6	85.404
Teacher	2488	12	.479*	.109	.849	2.537	241.823***	11	95.451
Both researcher and teacher	185	2	.618***	.423	.814	6.196	1.225	1	18.378
Self-administered	733	7	.623***	.331	.915	4.185	60.952***	6	90.156
Within levels							345.106***	24	
Between levels ^a							1.589/25.858***	3	
Overall	4373	28	.560***	.428	.692	8.303	370.964***	27	92.722

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 11
Educational Level

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
K-12	2237	15	.606***	.336	.876	4.399	204.003***	14	92.722
College	2025	11	.309*	.053	.565	2.369	133.504***	10	92.510
Adult	111	2	1.024***	.628	1.420	5.068	0.000	1	0.000
Within levels							337.644***	25	
Between levels ^a							9.071*/33.320***	2	
Overall	4373	28	.530***	.361	.700	6.121	370.964***	27	92.722

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

& Bohl, 2018; Muis et al., 2006). We found similar results across academic achievement measures at various levels of specificity, with some indication of a stronger effect on domain-general and topic-specific measures, compared to domain-specific. These findings are somewhat difficult to parse, as is the finding that studies with a mismatch in the levels of specificity of the epistemic cognition intervention and academic achievement measure were associated with higher effect sizes than studies that matched these levels, though this might simply be an artifact of the small number of cases that were not aligned. Regardless, we suggest that scholars target the level of specificity of their intervention instruction to the same level as their achievement measure, as most seem to be doing already.

The type of achievement results met our expectations to a large extent. Those studies that measured achievement via argumentation had the largest average effect size, which matches much of the literature on the importance of adaptive epistemic cognition when engaging in argumentation (e.g., Kuhn & Udell, 2003; Mason & Scirica, 2006). It is important to note that argumentation as instructional framework, discussed above, was not always aligned with the type of achievement. That is, in some studies students were instructed using argumentation-based interventions but then were assessed with an academic outcome that was categorized as conceptual knowledge (e.g., Lin, Chan, & van Aalst, 2014) or general knowledge (e.g., Osborne et al., 2013). This mismatch might explain why the instructional frameworks based on argumentation had statistically nonsignificant effects on achievement.

Past research has also demonstrated the importance of epistemic cognition on the acquisition of conceptual knowledge (e.g., Gill et al., 2004), and this relationship has been a focus of much recent scholarship (e.g., Amin & Levrini, 2018). The results of the moderator analysis indicated support for the idea that epistemic cognition instruction can bolster conceptual knowledge acquisition. Declarative knowledge measures also had a statistically significant, if lower, effect size, whereas studies involving measures of procedural or general (i.e. mixed) knowledge did not have statistically significant effect size estimates. Further, as might be expected, studies involving a researcher-designed achievement measure had a higher effect size estimate than studies involving measures developed by others. It is unclear whether this finding is due to greater sensitivity to changes in epistemic cognition among researcher-developed measures or potential biasing of the measure to match the intervention. Nonetheless, at some point the effect of epistemic cognition interventions on more typical achievement measures (e.g., course grades, GPA) must be established.

The effect sizes at all levels of schooling were statistically significant. Interventions enacted in K-12 settings yielded an effect size twice that of those carried out with college students. Some researchers in the field (e.g., Perry, 1970) have speculated that epistemic cognition may not develop much until adulthood, but the meta-analytic results demonstrate that scholars who have been working with younger children have strong justification to do so. The statistically significant results at all levels indicate that epistemic interventions can have positive effects throughout the life

Table 12
Length of Intervention

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Single Session ^a	336	4	.734***	.415	1.054	4.505	16.770***	3	82.111
Short-term	329	3	.615***	.364	.867	4.796	3.879	2	48.435
Intermediate	673	6	.925***	.333	1.517	3.064	120.458***	5	95.849
Long-term	348	4	.390	-.006	.786	1.930	18.854***	3	84.088
Semester (or longer)	2687	11	.234	-.034	.502	1.715	120.458***	10	92.681
Within levels							296.594***	23	
Between levels ^a							8.863/74.370***	4	
Overall	4373	28	.519***	.376	.662	7.106	370.964***	27	92.722

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 13
Time of Posttest Achievement Measure

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Immediate	4191	26	.451***	.267	.635	4.804	297.524***	25	91.597
Delayed	485	5	.966***	.802	1.131	11.503	5.262	4	23.979
Within levels							302.785***	29	
Between levels ^a							16.737***/78.924***	1	
Overall	4676	31 ^b	.737***	.615	.860	11.775	381.709***	30	92.141

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

^b Several studies (Barzilai & Ka'adan, 2017; Hefter et al., 2015; Huang, Ge, & Eseryel, 2017) used both immediate tests and delayed tests to measure achievement. The related effect sizes had to be separated to conduct the moderator analysis. This led to those participants in the studies being counted twice as unique.

* $p < .05$. ** $p < .01$. *** $p < .001$.

span, and researchers should continue to investigate these interventions with diverse sets of learners, including adults not in college (Greene et al., 2008).

Contrary to our expectations, the shorter interventions had a larger average effect than longer ones, with studies falling in our single session, short-term, and intermediate categories all resulting in statistically significant effect sizes. Studies in the longer categories, including the 11 studies in the “semester (or longer)” category were not statistically significantly associated with achievement gains. This seems peculiar, as Bendixen (2016) and others (e.g., VanSledright, 2002) have emphasized that effective epistemic cognition interventions must be enacted over time and with repeated exposure. However, similar findings have been shown in studies of reading (Wanzek et al., 2016) and mathematics interventions (Bailey et al., 2016). We might postulate, based on nonlinear growth modeling studies (e.g., Cameron, Grimm, Steele, Castro-Schilo, & Grissmer, 2015; Dumas & McNeish, 2017) that these longer interventions initially facilitate students' learning, but that this effect levels off when the students reach the asymptotic effectiveness of the intervention. After they level off, the longer the intervention goes, the more time the comparison or control group students have to catch up, thus reducing the effect sizes. This is largely speculative, though, as many of the studies did not include specifics about the total hours of intervention. Alternatively, for this particular meta-analysis, the achievement measures for the longer interventions in the meta-analysis were more likely to be standardized tests or teacher-developed assessments (see Table 16). It may be that measures more closely aligned with the

content of the interventions would have been more sensitive to effects.

Studies with delayed posttests produced an effect size estimate that was almost twice that of studies with an immediate measure. However, there were only five studies that had delayed measures, so this difference might be an artifact of a small sample size. It is also the case that three of the studies that had delayed measures also had immediate measures (Barzilai & Ka'adan, 2017; Hefter et al., 2015; Huang et al., 2017), and if participants received feedback after their immediate testing, then that might account for the higher scores on delayed tests, as perhaps those in the intervention groups were consequently able to use that feedback to develop their epistemic cognition on their own, over time. Another possibility, as Barzilai and Ka'adan (2017) proposed, is that the effects of an epistemic cognition intervention may take time to internalize and change beliefs, behaviors, and performance. Clearly, more research is needed on the long-term effects of epistemic cognition interventions after the treatment period ends.

As Bråten (2016) pointed out recently, the instructional frameworks that exist in epistemic cognition research “describe very complex, multi-componential approaches to teaching for epistemic change that are hardly amenable to deriving specific testable hypotheses for experimental work” (p. 363). However, he also allowed that these instructional frameworks should “spark subsequent empirical work that has the potential to refine and specify those frameworks as well as the theoretical models” (Bråten, 2016, p. 364). The results presented in this meta-analysis suggest numerous promising directions for future research, primarily based upon

Table 14
Research Design Moderator Analysis

Mixed effects analysis	N	k	ES	95% CI		Z-value	Test of heterogeneity		
				Lower limit	Upper limit		Q	df	I ²
Experimental	1142	10	.465***	.270	.660	4.676	48.074***	9	81.279
Quasi-experimental with equated groups	289	4	.767***	.412	1.122	4.232	12.800**	3	76.563
Quasi-experimental	2942	14	.470**	.158	.781	2.955	281.864***	13	95.388
Within levels							342.739***	25	
Between levels ^a							2.260/28.225***	2	
Overall	4373	28	.520***	.370	.670	6.801	370.964***	27	92.722

^a Between levels Q value entry is the mixed effects value followed by the fixed effects value.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 15
Peer Review Status

Mixed effects analysis	<i>N</i>	<i>k</i>	<i>ES</i>	95% CI		<i>Z-value</i>	Test of heterogeneity		
				Lower limit	Upper limit		<i>Q</i>	<i>df</i>	<i>I</i> ²
Not peer reviewed	1282	5	.335**	.127	.542	3.161	17.874***	4	77.621
Peer reviewed	3091	23	.552***	.323	.780	4.728	350.720***	22	93.727
Within levels							368.594***	26	
Between levels ^a							1.895/2.370	1	
Overall	4373	28	.433***	.279	.586	5.518	370.964***	27	92.722

^a Between levels *Q* value entry is the mixed effects value followed by the fixed effects value.

* *p* < .05. ** *p* < .01. *** *p* < .001.

those moderator levels with the strongest relation with academic achievement gains.

Limitations

As with any meta-analysis, our study had several necessary limitations. We did detect evidence of bias in the published literature. This problem, along with some relatively small sample sizes for particular levels of our moderator analyses, qualify our findings. Given that epistemic cognition intervention research remains a relatively new field, small sample size issues are not surprising. They can and will be addressed as the researchers continue to develop and refine interventions in the future.

Our ability to investigate all of the various models in the literature was limited because we could not locate any experimental interventions using an epistemological resources (Elby & Hammer, 2010) or AIR model (Chinn, Rinehart, & Buckland, 2014) perspective. There is a clear need for such research. Additionally, because of the small number of studies employing certain other theoretical frameworks, we felt compelled to combine similar models for the moderator analysis. Small sample sizes for levels of a moderator threaten both the power to identify a statistically significant difference between that level of a moderator and another, and they also call into question the accuracy of the point estimate for the effect size. Therefore, we decided to combine studies that used related theoretical models (e.g., the Bråten et al. and Greene et al. models) so that the moderator analysis could be conducted. Some such combinations were warranted given that authors often described their theoretical model as an extension or elaboration of another model (e.g., Barzilai and Zohar [2014], when describing their model, cited Kuhn et al. [2000] as a foundation for their work).

Other combinations were more challenging to make as the authors did not make such overt claims (e.g., combining De Corte et al.'s [2002] model from Mason and Scrivani's [2004] study with Hofer and Pintrich's [1997] model). Given such decisions to categorize studies with a combined moderator label can be questioned, scholars interested in interpreting the theoretical model moderator effects should review our coding and the original studies to determine if they agree with our decisions.

Similar concerns exist regarding our combinations of studies according to their instructional frameworks. Even studies with explicit similarities regarding the basis of their instruction (e.g., the argumentation instruction employed in Nussbaum et al. [2008] and Osborne et al. [2013]) had important differences in their aims and execution. Scholars might classify or interpret those differences in other ways than we did, particularly for those studies that were less explicit in the type of instructional framework employed as well as for those that did not clearly map onto frameworks shared by other studies (e.g., Olive-Taylor, 2007). Therefore, researchers interested in developing and deploying epistemic cognition interventions should review our decisions regarding how we coded the instructional frameworks in the individual studies included in this meta-analysis.

More generally, despite our best efforts to include all of the appropriate empirical research, we may have missed studies, in part due to the diverse terminology used in the field (e.g., personal epistemology, epistemic beliefs, epistemic cognition, etc.; Greene et al., 2016c). It is also possible that we included studies that were better left out. For example, there has been a great deal of scholarly debate about the utility and appropriateness of epistemic beliefs (e.g., Sandoval, 2009; Sinatra, 2016), as well as whether epistemic

Table 16
Type of Achievement Measure by Length of the Intervention

Length of the intervention	Type of the achievement measure				
	Grade point average	Researcher-developed measure	Standardized test	Teacher-developed measure	Grand total
Single session		13			13
Short-term		7			7
Intermediate		14			14
Long-term		5	1	4	10
Semester (or longer)	1	6	3	5	15
Overall	1	45	4	9	59

beliefs and epistemic cognition are related but somewhat separate phenomena (Sinatra, 2016). Such distinctions, if found to be important, may necessitate further research into how to account for both in intervention research.

Additionally, whether all of these interventions truly targeted epistemic cognition could be debated. As Greene and colleagues (2018) noted, some researchers self-identify as conducting epistemic cognition research but otherwise do not align their work with the underlying models or tenets of that literature. Other researchers might use terms that are atypical in epistemic cognition literature (e.g., constructivist instruction) but nonetheless develop an intervention clearly aligned with one or more models of epistemic cognition. Thus, there is inherent subjectivity in deciding which studies should be included in a meta-analysis of epistemic cognition interventions and which should not. For example, Muis et al. (2016) identified Higgins (1997) as an epistemic cognition intervention study. Higgins' (1997) study was similar to several studies we included in this meta-analysis in that constructivist teaching practices were employed, but we did not include Higgins's (1997) in this work because she framed her study as being about general beliefs about mathematical problem-solving rather than epistemic beliefs.

This issue is also critical when considering whether to include intervention studies on Nature of Science (NOS) understanding. Muis et al. (2016) identified several NOS interventions (e.g., Akerson & Hanuscin, 2007; Akerson, Morrison, & McDuffie, 2006) as empirical studies on epistemic change, and there are other similar such studies (e.g., Billeh & Hasan, 1975). NOS is indeed primarily concerned with epistemological ideas, and some scholars have pointed out important overlaps in the fields of epistemic cognition and NOS (e.g., the emphasis on the tentative nature of knowledge; Cho, Lankford, & Wescott, 2011), but there are also areas where they do not map onto each other (e.g., NOS has a factor related to the "myth of the scientific method"; Abd-El-Khalick, 2012). Given that we are aware of no other meta-analysis of epistemic cognition interventions, we chose to narrowly define the sample, and we excluded NOS interventions.

Including studies like Higgins' (1997) and NOS studies similar to those discussed above ultimately would have added to the overall power of the analysis. This boost in power would have consequently increased the likelihood, especially in the moderator analyses, of finding statistically significant differences between groups. However, we believe that the interpretability of the results would have been compromised by including studies that did not map clearly onto the epistemic cognition focus of the meta-analysis. Thus, though we recognize that the small number of included studies means that the overall analysis and the moderator analyses are underpowered, we also believe that restricting the inclusion of studies to ones that clearly fit our criteria provides the most transparent interpretations possible. Given this limitation, researchers and educators should interpret the moderator analysis findings with the understanding that other differences across levels of moderators may exist in the population but may not be detectable given the size of the current literature. Our findings suggest promising directions for the development of future interventions, but that does not mean that theoretically supported directions that did not show statistically detectable differences in this meta-

analysis (e.g., interventions with a problem-based learning instructional framework) should be completely abandoned.

Future Directions

Regardless of the limitations, though, the findings in this meta-analysis do indicate that interventions based on improving learners' epistemic cognition can have positive effects on academic achievement. In general, there seem to be some basic practices that should be used in future interventions along these lines. These practices include using guided forms of instruction, focusing on justification norms within disciplines, utilizing achievement measures targeting higher-order thinking such as argumentation or conceptual knowledge acquisition, and conducting interventions at all educational levels. Another route that researchers might take is planning more interventions focusing on source evaluation. With the emphasis on justification in several prominent epistemic cognition models (Bråten et al., 2011; Greene et al., 2008) and the large effect sizes achieved by the two studies that employed source evaluation instruction that were included in this analysis, it seems like that line of study could be ripe for extended investigation. Due to the small number of studies in each case, we cannot say that epistemic cognition interventions based on problem-based learning and argumentation frameworks, which yielded nonstatistically significant results, should be avoided. Indeed, the close relationship between epistemic cognition and argumentation, in particular, and the statistically significant effects of epistemic cognition interventions in general upon argumentation assessments, suggest that instruction on argumentation and epistemic cognition may be fruitful. Further, researchers planning interventions based on these types of frameworks should carefully consider matching their achievement outcomes to their instructional goals (i.e. an argumentation-based intervention should be evaluated based on an argumentation-based assessment).

The results of this meta-analysis suggest several avenues for future research. Most epistemic cognition research has been conducted with college students, but our analysis suggests that K-12 students benefit just as much, if not more, from epistemic cognition interventions. Greater use of immediate and delayed posttests would likely clarify whether these interventions have lasting effects. Researchers should work to examine interventions in disciplines beyond science, and more research on topic-specific interventions is needed. Only two of the studies included in the meta-analysis were conducted in laboratory settings. We recognize the necessity of having interventions that work in classrooms, nonetheless it may be helpful for the sake of fine-tuning and identification of active ingredients for some experiments to be conducted in a more controlled laboratory setting.

Longer interventions, usually lasting approximately a semester, make up a large portion of the studies in the literature, yet we found shorter interventions yielded better results. This may be because the achievement measures used in shorter interventions better fit the target of the interventions, or it may be because the concentration of the epistemic cognition instruction is greater in brief studies. It may also be because of issues regarding fadeout. We join Wanzek et al. (2016) and others in encouraging researchers to provide more specific and precise

information about dosages in educational interventions. Researchers should also consider how to account for confounding factors between longer interventions and development based on maturity and experiences.

Finally, one important change to typical practice would be for researchers to measure or gauge the participants' epistemic cognition during the study. Surprisingly, several studies included in this meta-analysis (e.g., Kim & Keller, 2010; Kuhn et al., 2017) never measured this or did not report doing so. It would also be a best practice to conduct these evaluations before and after the intervention to see if the students' epistemic cognition did actually change. The reluctance to conduct this kind of assessment may lie in the nature of the self-report instruments that tend to be used, which have been strongly criticized (e.g., DeBacker, Crowson, Beesley, Thoma, & Hestevold, 2008; Mason, 2016). We are not advocating that these instruments be used, or at least not exclusively. There are other techniques like cognitive interviews (Greene & Yu, 2014), think-aloud protocols (Hofer, 2004), and artifact production and discourse (Sandoval, 2009) that could be employed, and the selection of the measure should depend largely on the instructional framework (Bråten, 2016). Gathering such data will certainly increase the time and effort involved in this aspect of the interventions, but the information gleaned will be critical in improving future instruction and for continuing to establish the validity, or lack thereof, of the various measures.

Conclusion

In sum, this meta-analytic review of experimental and quasi-experimental epistemic cognition intervention studies and their effects on academic achievement revealed a medium effect that supports further development and refinement of these interventions. Our moderator analysis findings suggested promising directions for future development (e.g., guided inquiry instruction, a focus on justification and source evaluation, interventions at the K-12 level) as well as a need for additional research in particular disciplines (e.g., history and the literary arts). Further, researchers should strongly consider measuring epistemic cognition both before and after the intervention as a way of demonstrating that the active ingredient in the intervention is indeed epistemic cognition. With continued improvements and studies, research on epistemic cognition interventions should contribute greatly to better preparing students for the higher-order thinking that is critical to effective learning and functioning in the 21st century.

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